Graph Neural Networks for Node Classification in Citation Networks: A Comparison of GCN and GAT

Abstract

Graph Neural Networks (GNNs) are making waves in the field of graph-structured data, like social networks, citation networks, and biological systems. In citation networks, a node stands for a research paper, and an edge connects papers through citation relationships. This project takes a close look at two popular GNN architectures: Graph Convolutional Networks (GCN) and Graph Attention Networks (GAT). The aim? To see how they stack up in node classification tasks using citation network datasets. The experiments involved the Cora dataset and the large-scale OGBN-Arxiv dataset. The results are interesting—GAT edges out GCN in classification accuracy on the Cora dataset, showing that attention mechanisms have advantages on smaller graphs. But when it comes to the much larger OGBN-Arxiv dataset, GCN delivers better performance than GAT in the tested setup. The takeaway is clear: attention mechanisms boost performance on smaller graphs, but when the graph grows, simpler convolution-based models seem just as strong, if not more reliable.

1. Introduction

Many real-world datasets can naturally be represented as graphs. In a graph structure, nodes represent entities while edges represent relationships between them. Examples include social networks, transportation networks, biological interaction networks, and citation networks. Traditional machine learning models

often assume that data samples are independent of each other, which limits their ability to capture the relational structure present in graph data.

Graph Neural Networks (GNNs) have been developed to address this limitation by learning representations directly from graph structures. These models aggregate information from neighboring nodes in order to learn meaningful node embeddings. As a result, GNNs have become widely used for tasks such as node classification, link prediction, and graph classification.

Among the many GNN architectures proposed in recent years, Graph Convolutional Networks (GCN) and Graph Attention Networks (GAT) are two of the most influential models. GCN performs neighborhood aggregation through graph convolution operations, allowing each node to incorporate information from its neighbors. GAT extends this idea by introducing an attention mechanism that assigns different importance weights to neighboring nodes.

The goal of this project is to compare the performance of GCN and GAT on citation network datasets. Specifically, the models are evaluated on the Cora dataset and the OGBN-Arxiv dataset. By analyzing the performance of both models across different graph sizes, this study aims to understand how attention mechanisms influence node classification performance in graph neural networks.

2. Related Work

Graph Neural Networks have gained significant attention in recent years due to their ability to effectively model relational data. One of the most influential models in this field is the Graph Convolutional Network (GCN) proposed by Kipf and Welling (2017). GCN applies convolution operations to graph data by aggregating

feature information from neighboring nodes using a normalized adjacency matrix. This approach allows the model to learn node representations that capture both node features and graph structure.

Building on this idea, Veličković et al. (2018) introduced Graph Attention Networks (GAT). Unlike GCN, which treats all neighbors equally during aggregation, GAT uses an attention mechanism to assign different weights to neighboring nodes. This enables the model to focus more on relevant neighbors while reducing the influence of less informative nodes.

In addition to model architectures, benchmark datasets have played an important role in evaluating graph learning algorithms. Citation network datasets such as Cora, Citeseer, and PubMed have been widely used in early graph learning research. More recently, the Open Graph Benchmark (OGB) introduced large-scale datasets for evaluating graph machine learning models. The OGBN-Arxiv dataset is a commonly used benchmark for node classification tasks in citation networks.

Previous studies have demonstrated strong performance of both GCN and GAT on small citation datasets. However, their performance on large-scale graphs varies depending on model complexity, training configuration, and dataset characteristics.

3. Proposed Methodology

This project compares the performance of Graph Convolutional Networks (GCN) and Graph Attention Networks (GAT) for node classification tasks on citation network datasets.

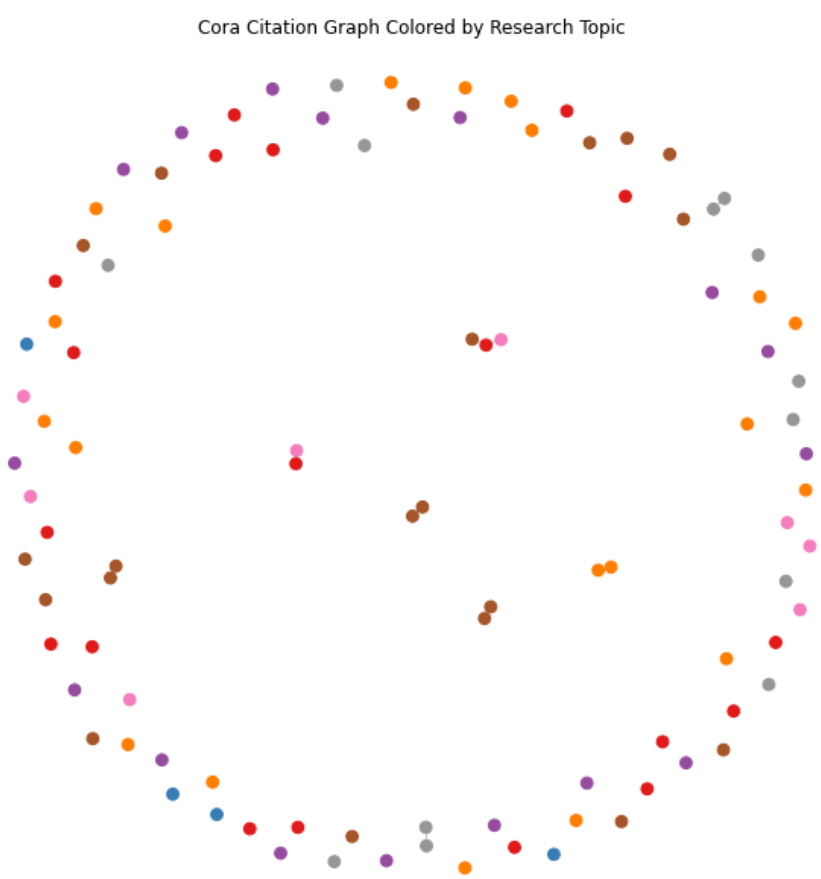
Datasets

Two citation network datasets were used in the experiments:

Cora Dataset

The Cora dataset is a commonly used benchmark for graph learning research. It contains:

- 2,708 nodes (scientific publications)
- 5,429 edges (citation relationships)
- 7 classes representing research topics
- 1,433-dimensional node features



Each node represents a research paper, and edges represent citations between papers.

OGBN-Arxiv Dataset

The OGBN-Arxiv dataset is a large-scale citation network provided by the Open Graph Benchmark. It contains:

- 169,343 nodes
- Over 1.1 million edges
- 40 research categories
- 128-dimensional node features

This dataset represents a significantly larger and more complex graph compared to Cora.

Graph Convolutional Network (GCN)

The GCN model used in this project consists of two graph convolution layers.

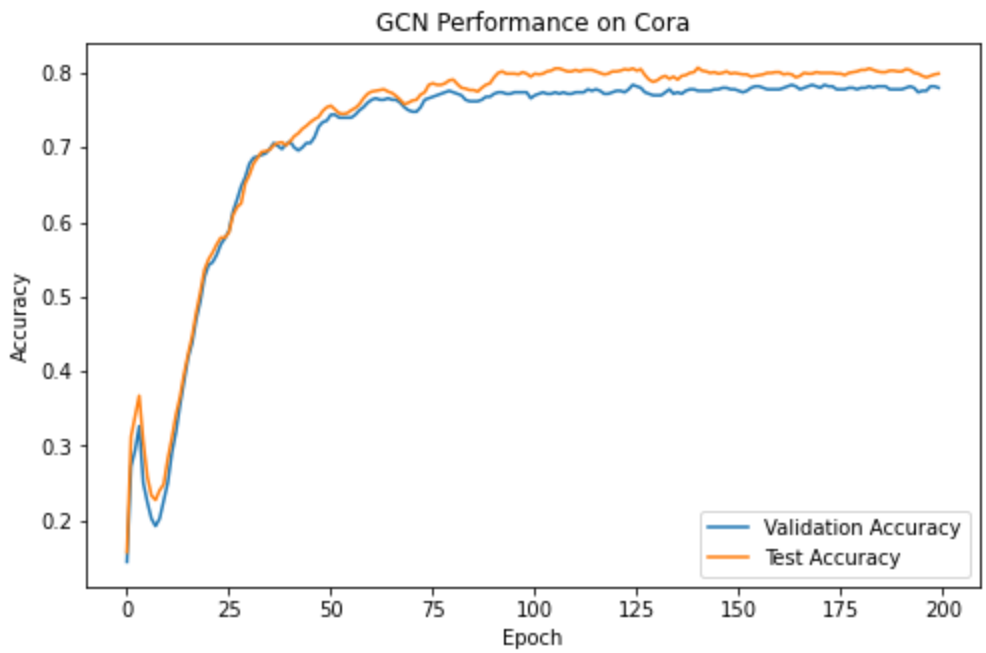
1. The first GCN layer aggregates feature information from neighboring nodes.

2. A ReLU activation function introduces non-linearity.

3. Dropout regularization is applied to reduce overfitting.

4. A second GCN layer produces the final node classification outputs.

The model learns node embeddings by iteratively aggregating features from neighboring nodes in the graph.



Graph Attention Network (GAT)

The GAT model introduces an attention mechanism during the aggregation process. Instead of treating all neighboring nodes equally, GAT learns attention weights that indicate the importance of each neighbor.

The architecture used in this project includes:

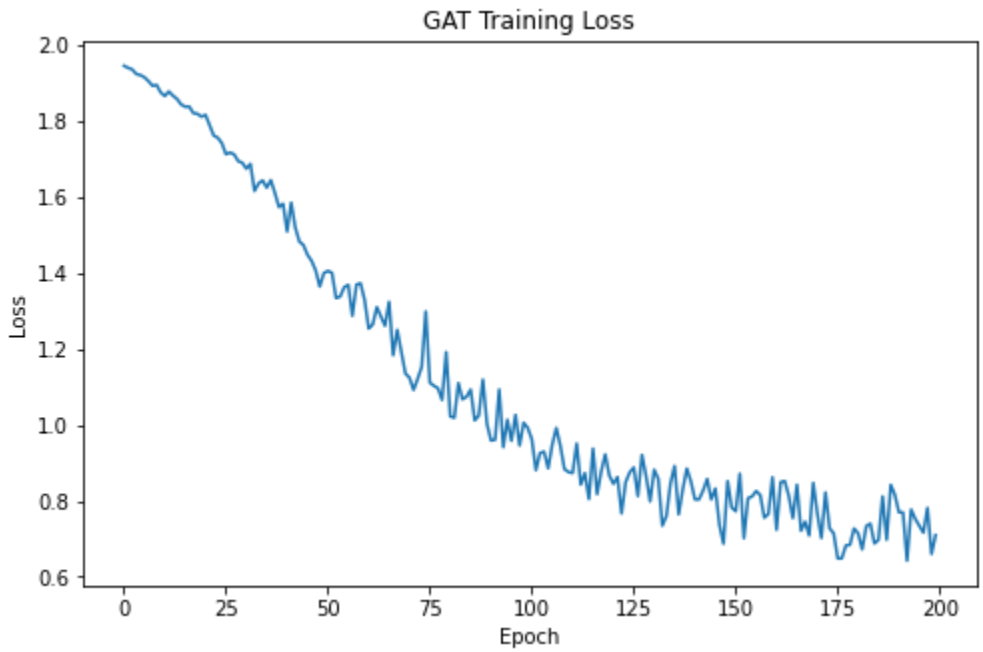
1. A first GAT layer with multiple attention heads.

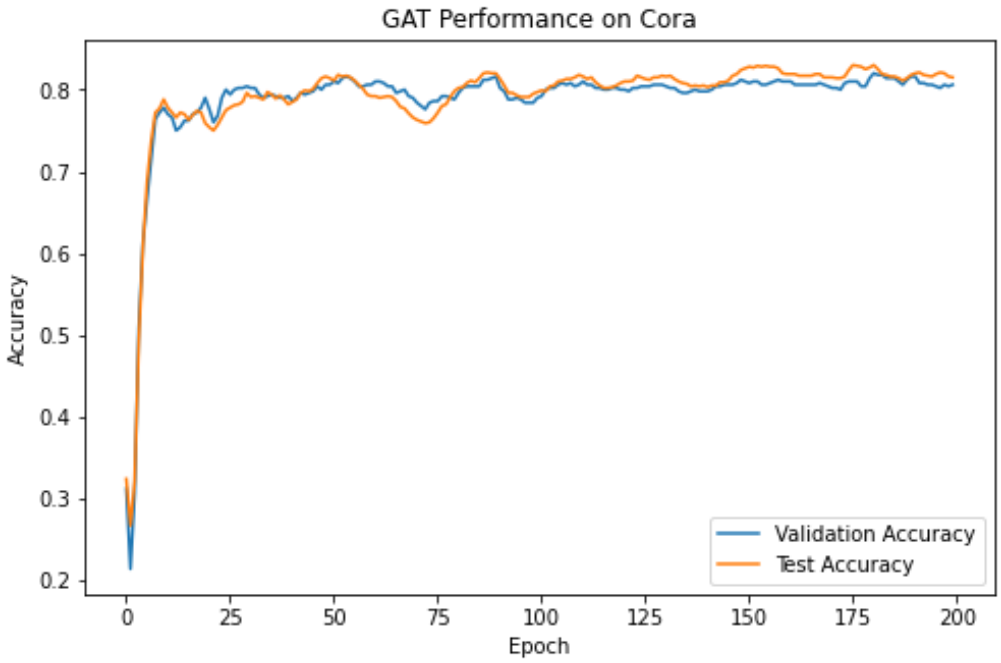
2. An ELU activation function.

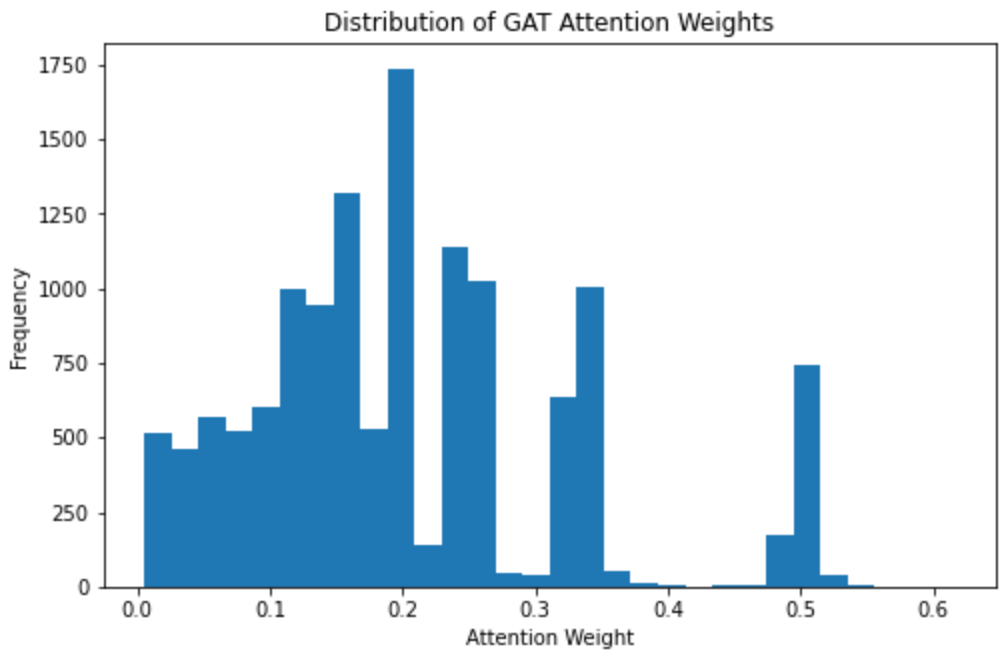
3. Dropout regularization.

4. A final GAT layer for classification.

The attention mechanism allows the model to selectively focus on important neighbors when computing node representations.







Training Procedure

Both models were implemented using PyTorch and PyTorch Geometric. The models were trained using the Adam optimizer with cross-entropy loss for node classification.

For the Cora dataset, the standard dataset masks were used to split nodes into training, validation, and test sets.

For the OGBN-Arxiv dataset, the official dataset splits provided by the Open Graph Benchmark were used.

Each model was trained for 100 epochs, and the best model was selected based on validation accuracy.

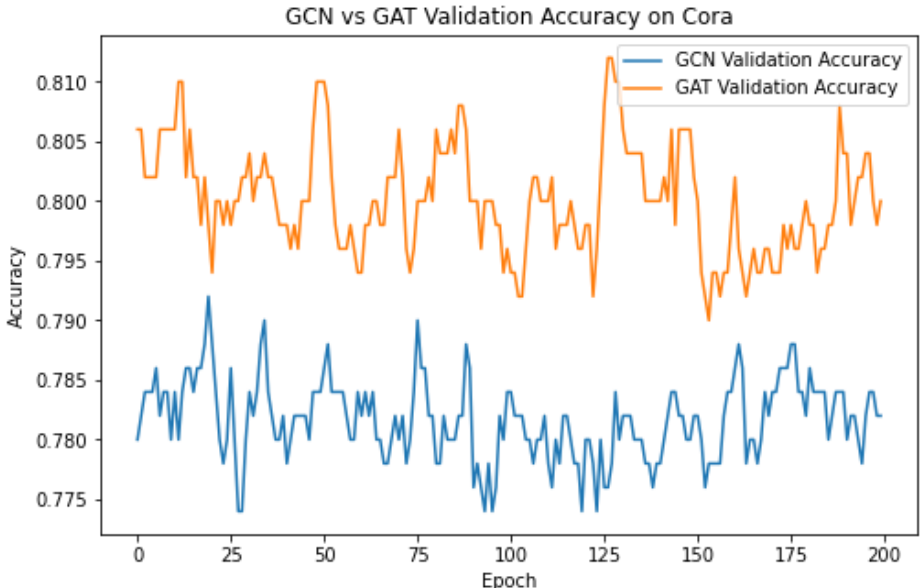
4. Evaluation and Discussion

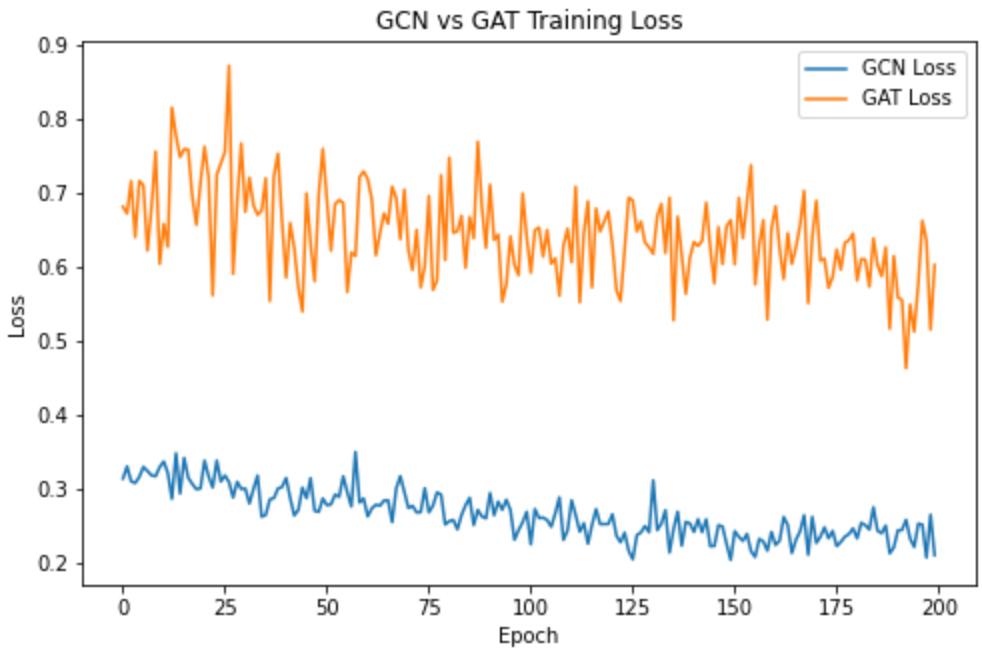
The models were evaluated using validation accuracy and test accuracy. For the Cora dataset, macro F1-score was also calculated to evaluate classification performance across all classes.

Results on Cora

Model	Validation Accuracy	Test Accuracy	Test Macro F1
GCN	0.794	0.804	0.793
GAT	0.812	0.820	0.810

The results indicate that GAT performs slightly better than GCN on the Cora dataset. The attention mechanism allows the model to learn the relative importance of neighboring nodes, which improves classification performance in smaller graphs.



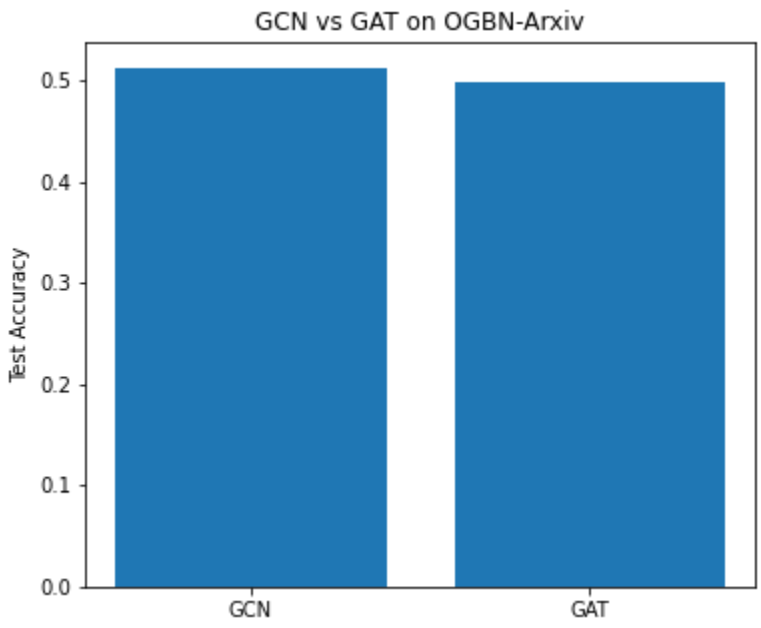


Results on OGBN-Arxiv

Model	Validation Accuracy	Test Accuracy
GCN	0.574	0.512
GAT	0.539	0.497

On the larger OGBN-Arxiv dataset, the GCN model performs better than GAT. This may be due to the increased complexity of the dataset and the computational overhead of attention mechanisms. Attention-based models often require additional hyperparameter tuning and training adjustments to achieve optimal performance on large graphs.

Overall, the results suggest that while GAT can improve performance on smaller citation networks, GCN remains a strong and stable baseline model for large-scale graph datasets.

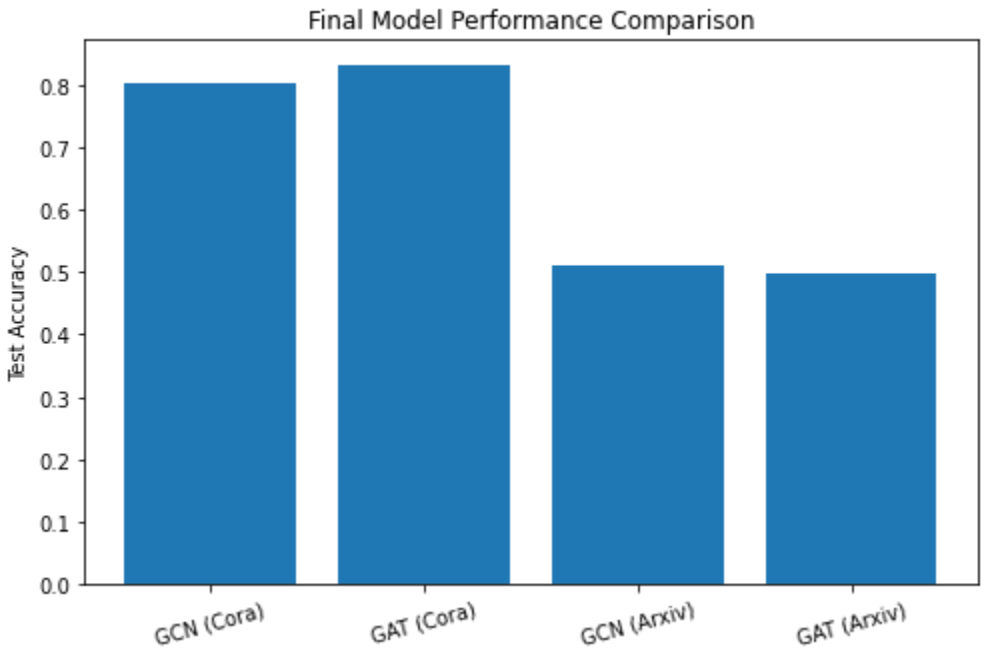


5. Conclusion

This project compared Graph Convolutional Networks and Graph Attention Networks for node classification tasks in citation networks. Experiments were conducted using the Cora dataset and the OGBN-Arxiv dataset.

The results show that GAT achieves higher classification accuracy than GCN on the Cora dataset, demonstrating the advantages of attention-based aggregation in smaller graphs. However, on the larger OGBN-Arxiv dataset, GCN achieved better performance under the current experimental setup.

These findings indicate that the effectiveness of attention-based graph neural networks may depend on dataset size, graph complexity, and model configuration. Future work could explore improved GAT architectures, additional hyperparameter tuning, or scalable attention mechanisms for large graphs.



References

Kipf, T. N., & Welling, M. (2017).
Semi-Supervised Classification with Graph Convolutional Networks.
https://arxiv.org/abs/1609.02907

Veličković, P., Cucurull, G., Casanova, A., Romero, A., Liò, P., & Bengio, Y. (2018).
Graph Attention Networks.
https://arxiv.org/abs/1710.10903

Hu, W., Fey, M., Zitnik, M., Dong, Y., Ren, H., Liu, B., Catasta, M., & Leskovec, J. (2020).
Open Graph Benchmark: Datasets for Machine Learning on Graphs.

Project Github link:

https://github.com/dumbyoshi/Graph-Neural-Networks-GNNs-.git